

Assignment #3: Anonymization of the *Adult Dataset* with Risk and Utility Analysis

Security and Privacy — FCUP 2025/2026

Francisco Machado (up202403514)

Rodrigo Dias (up202407281)

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1 Introduction

This work performs the anonymization of the *Adult dataset* (Census Income, UCI), comprising **32,561 records** and 15 attributes, using the ARX 3.9.2 tool. The goal is to classify the attributes according to their role in privacy, characterize the re-identification risk of the original dataset, apply two privacy models (k -anonymity and k -anonymity + ℓ -diversity), and analyze the trade-off between risk and utility as a function of the privacy models' parameters.

2 Attribute Classification

Classification was based on the **distinction** metric (percentage of distinct values, closeness to a key) and **separation** (percentage of record pairs the attribute can distinguish), as analyzed in ARX's *Quasi-identifiers* tab. Table 1 summarizes the values and the assigned classification.

Attribute	Dist.%	Sep.%	Class	Justification
fnlwgt	66,48	99,99	Identifying	Near-unique distinction; sampling weight with no analytical value.
education-num	0,05	80,96	Identifying	Exact numeric duplicate of <i>education</i> - removed to avoid duplicating the dimension.
age	0,22	97,87	QID	Very high separation; age is known/inferable and highly discriminative.
occupation	0,05	90,29	QID	High separation; often public (e.g. LinkedIn).
education	0,05	80,96	QID	Known demographic, high separation.
hours-per-week	0,29	76,25	QID	Observable/inferable, high separation.
marital-status	0,02	66,01	QID	Public demographic; contributes to the identifying combination.
workclass	0,03	49,71	QID	Observable demographic.
sex	0,01	44,28	QID	Publicly known; a multiplier within the combination.
race	0,02	25,98	QID	Visible/known demographic attribute.
native-country	0,13	19,66	QID	Low separation (90% US), but rare values make individuals unique.
income	0,01	36,57	Sensitive	Target attribute, financially sensitive; protected via ℓ -diversity.
capital-gain	0,37	15,93	Sensitive	Private financial data to protect against inference.
capital-loss	0,28	9,10	Sensitive	Idem: private financial information.
relationship	0,02	73,21	Insensitive	Redundant with <i>marital-status/sex</i> ; kept without transformation.

Table 1: Classification of the 15 attributes based on distinction and separation.

Justification notes. (i) *fnlwgt* and *education-num* are both *Identifying* but for different reasons: the former due to near-unique distinction (66%), the latter because it is an exact copy of *education* (avoids counting the same dimension twice among the QIDs). (ii) *income* is *Sensitive* rather than a QID because it is the target of inference: protecting it is the goal of ℓ -diversity; being **binary** ($\leq 50K$ / $> 50K$), it limits the maximum value of ℓ to 2. (iii) *relationship* is *Insensitive* as it correlates strongly with *marital-status* and *sex*: classifying it as a QID would add redundancy with no privacy gain. (iv) *capital-gain* and *capital-loss* are *conceptually* sensitive (private financial data), but were declared *Insensitive* in ARX for a practical reason: with $> 91\%$ of their values equal to 0, most equivalence classes contain a single distinct value, so a *distinct* ℓ -diversity criterion ($\ell \geq 2$) on them is infeasible: it admits no anonymous solution within any reasonable suppression limit, which prevents the models from running. Since ARX does **not** generalize sensitive attributes (it only

Distribution of risks	Quasi-identifiers	Attacker models	HIPAA identifiers
Quasi-identifier	Distinction	Separation	
sex	0.00614%	44.27526%	
race	0.01536%	25.9841%	
relationship	0.01843%	73.21449%	
marital-status	0.0215%	66.01289%	
workclass	0.02764%	49.71296%	
occupation	0.04607%	90.28866%	
education-num	0.04914%	80.96047%	
education	0.04914%	80.96047%	
age	0.22419%	97.86783%	
fnlwtg	66.48444%	99.9962%	

Figure 1: ARX *Quasi-identifiers* tab showing the distinction and separation values used to classify the attributes (cf. Table 1).

carries them through unchanged), declaring them *Insensitive* produces the same untransformed output while letting the k -anonymity and $k + \ell$ models execute.

Risk arises from the combination. No single QID identifies an individual, but the combination of the 9 QIDs produces **26,536 equivalence classes**, of which **23,512 are unique records (72,21%)**. The combined distinction is 81,5%. It is this exposure that motivates the anonymization.

3 Coding Models: Generalization Hierarchies

A generalization hierarchy was defined for each QID. Numeric attributes used **interval**-based hierarchies; categorical ones were built by successively grouping values (*merge*). Table 2 summarizes the structure.

QID	Type	Levels	Structure (from base level to top)
age	Intervals	5	exact $\rightarrow$ 5 years $\rightarrow$ 10 years $\rightarrow$ 20 years $\rightarrow$ *
hours-per-week	Intervals	4	exact $\rightarrow$ blocks of 10 $\rightarrow$ 2 bands (part/full-time) $\rightarrow$ *
education	Categorical	4	16 values $\rightarrow$ {Without-HS, HS, Higher-Ed, Post-Grad} $\rightarrow$ {Basic, Advanced} $\rightarrow$ *
marital-status	Categorical	4	7 values $\rightarrow$ {Married, Was-married, Single} $\rightarrow$ {Has-been-married, Single} $\rightarrow$ *
workclass	Categorical	3	8 values+? $\rightarrow$ {Government, Self-emp, Private, Unemployed} $\rightarrow$ *
occupation	Categorical	3	14 values+? $\rightarrow$ {White-collar, Blue-collar, Service, Other} $\rightarrow$ *
race	Categorical	3	5 values $\rightarrow$ {White, Non-White} $\rightarrow$ *
sex	Categorical	2	{Male, Female} $\rightarrow$ *
native-country	Categorical	3	41 values+? $\rightarrow$ {North-America, Latin-America, Europe, Asia} $\rightarrow$ *

Table 2: Generalization hierarchies defined for the 9 quasi-identifiers.

Relevant grouping decisions.

- **age vs hours-per-week.** Interval width does not follow a formula: for *age*, with a spread distribution, fine 5-year intervals preserve useful information (age bands are analytically meaningful), justifying a deep hierarchy. For *hours-per-week*, where **46,7% of records are exactly 40h** and 56,3% fall in the 40–49h block, 5h blocks would be redundant; blocks of 10h were used, sufficient to distinguish part-time / full-time / overtime.
- **Dominant categories stay isolated.** In *race*, *White* (85,4%) and in *native-country*, *United-States* (~90%) are so common that they carry no risk; the *rare* categories (those that make individuals unique) are generalized instead. It is acknowledged as a **limitation** that grouping ethnic minorities into a single *Non-White* erases distinctions between groups: a pragmatic anonymization choice, acceptable given the objective.
- **Disambiguation of *native-country*.** The value *South* is not explicitly documented; we follow the common interpretation that it corresponds to *South Korea* and grouped it under Asia. This decision is supported by the data itself: **96,2% of records with South have race Asian-Pac-Islander**, a proportion consistent with (and higher than) that of the confirmed Asian countries. By the same truncation logic, *Hong* was treated as *Hong Kong* (Asia). Missing values (?) were mapped directly to *.

4 Risk of the Original Dataset

The risk analysis (*prosecutor* model, the most conservative attack) confirms the high exposure of the original dataset: a **maximum risk of 100%** (unique records re-identifiable with certainty) and an **average risk of 81,5%**. About **72,2%** of records are *sample uniques* and the percentage of *population uniques* (PITMAN model) is 13,9%. The average equivalence-class size is only 1,23 records. These values constitute the **baseline** for assessing the effect of anonymization.

5 Privacy Models: Risk and Utility

Two models were applied with **global transformation** and a **suppression limit of 100%**, using *Loss* as the utility metric (geometric mean, generalization/suppression factor = 0,5). Information loss is reported by the *Score* of the optimal solution (0 = no loss, 1 = total loss).

5.1 Model 1: k -anonymity

We varied $k \in \{2, 5, 10, 25, 50, 100\}$. Table 3 presents the results and Figure 2 the risk and utility plots as a function of k .

k	Max. risk (%)	Success rate (%)	Avg. class size	Suppression (%)	Loss (Score)
2	50,0	7,47	13,39	7,86	0,181
5	20,0	2,62	38,23	10,05	0,260
10	10,0	1,62	61,86	13,93	0,311
25	4,0	0,63	159,19	12,98	0,366
50	2,0	0,36	280,49	13,86	0,409
100	0,97	0,19	514,43	16,26	0,451
<i>Original (reference)</i>		81,50	1,23	0	—

Table 3: k -anonymity model: risk and utility as a function of k .

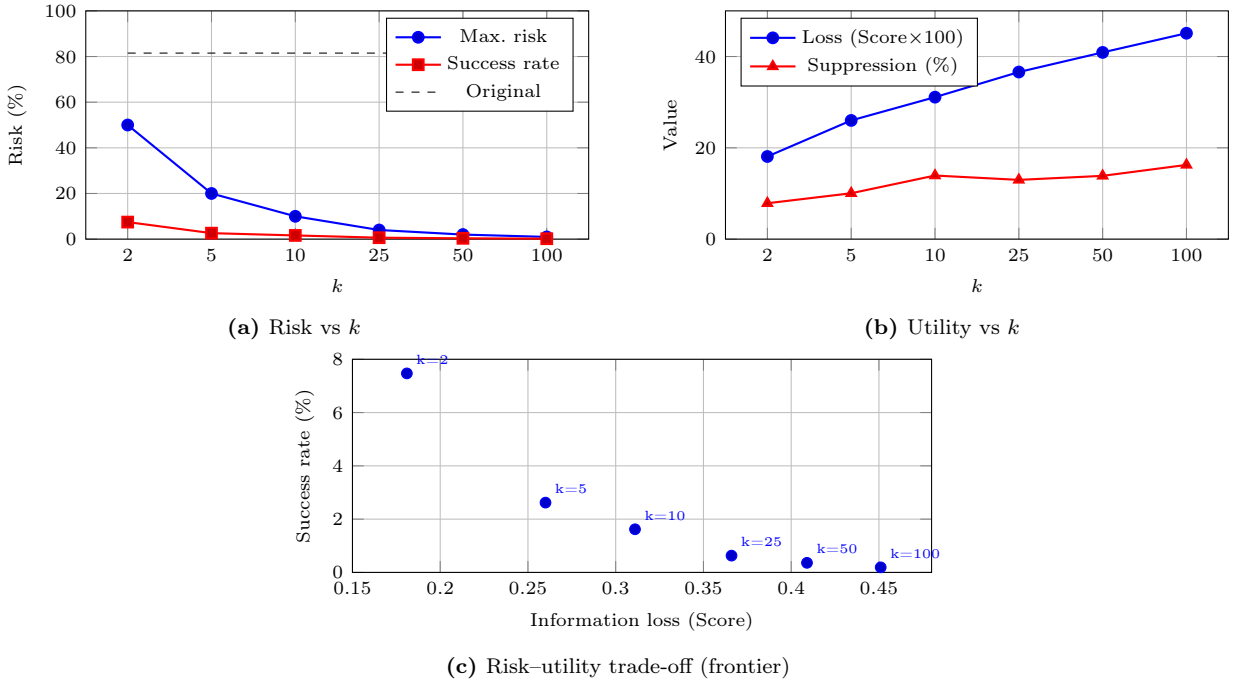


Figure 2: Risk and utility of the k -anonymity model as a function of k . The maximum risk follows $1/k$; utility degrades monotonically.

Analysis. Risk decreases predictably: the maximum risk follows $1/k$ (50%, 20%, 10%, 4%, 2%, $\sim 1\%$) and the *success rate* drops from 7,47% to 0,19%. Even $k = 2$ drastically reduces risk relative to the original’s 81,5%. In contrast, information loss grows monotonically (0,18 $\rightarrow$ 0,45). Notably, **suppression is not strictly monotonic** ($k = 25$ suppresses 12,98%, less than the 13,93% of $k = 10$): by generalizing one attribute more aggressively (e.g. raising *age* from level 3 to 4), ARX reduces the need to suppress records, illustrating the internal trade-off between generalization and suppression. The **“knee” of the curve** lies at $k = 5$ –10, where

low risk ($\leq 10\%$) is already achieved with still-moderate loss; beyond $k = 25$ the privacy gain is marginal but the utility cost is high.

5.2 Model 2: k -anonymity + ℓ -diversity

We added ℓ -diversity (*distinct*) over the sensitive attribute *income*, keeping $k = 5$. Since *income* is binary, $\ell = 2$ is the **maximum possible**: an inherent limitation of the attribute, not a configuration choice, so an ℓ sweep is not meaningful. ℓ -diversity was not applied to *capital-gain/capital-loss*: both have an extremely skewed distribution ($>91\%$ zero values), which makes *distinct* ℓ -diversity inadequate: a class with $\{0, 0, 0, X\}$ satisfies $\ell = 2$ but offers no real protection against inference. Table 4 compares the two models at $k = 5$.

Metric	Model 1 ($k=5$)	Model 2 ($k=5, \ell=2$)	Effect of ℓ -diversity
Maximum risk (%)	20,0	20,0	unchanged (k controls re-identification)
Success rate (%)	2,62	1,22	lower (-53%)
Avg. class size	38,23	81,94	more aggregated classes
Min. class size	5	5	unchanged
Suppression (%)	10,05	10,41	slightly higher
Loss (Score)	0,260	0,345	higher ($+33\%$)

Table 4: Comparison of pure k -anonymity with k -anonymity + ℓ -diversity ($k = 5$).

Analysis. ℓ -diversity **does not change the re-identification risk** (maximum risk stays at 20%, controlled by k), but it adds protection against a distinct attack vector - **sensitive attribute inference**: by requiring each equivalence class to contain both values of *income*, it prevents the attacker from deducing an individual’s income even without identifying them. The cost is higher information loss ($+33\%$) and larger classes ($38 \rightarrow 82$), since the additional constraint forces more generalization (*age* rises to level 4 and *marital-status* to level 3, which remained at 0 in Model 1). The *success rate* also drops, but as a side effect of the increased class size.

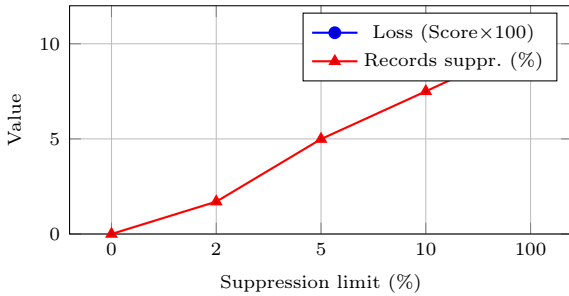
5.3 Generalization vs suppression

The optimal transformation vectors show that ARX favors generalizing *age* and *hours-per-week* (attributes with deep hierarchies), keeping *marital-status* and *sex* ungeneralized for low k . Suppression acts as a complementary mechanism to eliminate the *outliers* that would otherwise force the generalization of an entire column, hence the non-monotonic relationship observed in Table 3.

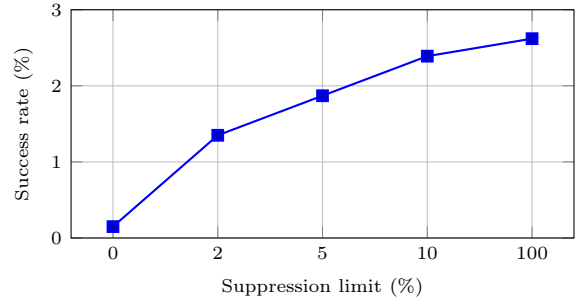
To isolate this trade-off, we performed a dedicated sweep of the **suppression limit** with $k = 5$ fixed (pure k -anonymity), testing the values $\{0, 2, 5, 10, 100\}\%$. Table 5 and Figure 3 present the results.

Suppr. limit (%)	Max. risk (%)	Success rate (%)	Min. class	Records suppr. (%)	Loss (Score)
0	7,14	0,15	14	0,00	0,346
2	20,0	1,35	5	1,70	0,282
5	20,0	1,87	5	4,99	0,264
10	20,0	2,39	5	7,50	0,262
100	20,0	2,62	5	10,05	0,261

Table 5: Suppression-limit sweep with $k = 5$ fixed. The 0% case cannot suppress any record and is forced to over-generalize.



(a) Information loss vs records suppressed



(b) Residual risk (success rate)

Figure 3: Effect of the suppression limit at $k = 5$. Allowing more suppression lowers generalization-driven information loss but raises both the suppression count and the residual risk, until both plateau near the $\sim 10\%$ the optimal solution naturally requires.

Analysis. The sweep cleanly exposes the generalization / suppression trade-off. The 0% **case is the most revealing**: forbidden from suppressing any record, ARX must generalize until *every* class (including outliers) reaches $k = 5$, producing the deepest transformation ($[3, 1, 2, 1, 1, 1, 0, 3, 2]$), the highest information loss (0,346), and — as a side effect — an average class size of 678 and a maximum risk of only **7,14%** (well below the requested 20%, since the minimum class ends up at 14). As the suppression limit increases, ARX trades generalization for the removal of those outliers: information loss drops ($0,346 \rightarrow 0,261$) while suppressed records rise ($0\% \rightarrow 10,05\%$) and the residual risk climbs toward the $1/k = 20\%$ ceiling. Both curves **plateau beyond $\sim 10\%$** , because the optimal solution never needs to suppress more than the 10,05% it requires at the 100% limit — so limits of 10% and 100% yield virtually identical results. The best balance lies around a **2–5% limit**, which recovers most of the utility lost at 0% while suppressing only a small fraction of records.

6 Iterations and Recommended Configuration

The k sweep, the suppression-limit sweep, and the introduction of ℓ -diversity constitute the planned refinement iterations: the first identifies the risk/utility equilibrium point, the second tunes the generalization/suppression balance, and the third adds protection against inference at a utility cost. As an additional refinement, **local transformation** (instead of global) could improve utility by allowing distinct generalization levels per group of records, a direction to explore in future work. **Recommended configuration:** for a balanced risk/utility trade-off, $k = 5$ to $k = 10$ with a suppression limit of about 2–5% (which recovers most of the utility while suppressing few records); if protection against *income* inference is a requirement, $k = 5 + \ell = 2$.

7 Conclusion

Anonymization reduced the re-identification risk drastically: the *success rate* fell from 81,5% (original) to 2,62% ($k = 5$) and the worst case from 100% to 20%, with 0% of records at high risk. The utility cost is controlled and grows with k . ℓ -diversity complements k -anonymity by covering a different attack vector. The risk/utility trade-off confirms $k = 5$ –10 as the best-balance region for this dataset.

Documentation of AI Use

In compliance with the course policy on academic integrity, the authors disclose that LLM assistants (Anthropic Claude and Google Gemini) were consulted in a limited and supervised manner for clarification of technical concepts, guidance on the ARX workflow and minor LaTeX formatting and drafting assistance. All anonymization runs, privacy-model configuration, metric collection, risk assessment, and validation of technical claims were performed exclusively by the authors, who take full responsibility for the content of this report.